

HomeWork1(CSCI-7350)

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- Problem 1 – Linear Regression Gradient Descent Algorithm
 1. Compute Loss with the current coefficients. Did loss increase or decrease?
 2. Compute the gradient vector $g=[g_0, g_1]$
 3. Update weights w_0 and w_1

All the calculations have been shown at the end of this document.

- Problem 2 – Linear Regression on Real Dataset
 1. Show the complete linear regression model learned in the form of: $\text{target} = w_0 + w_1 * \text{feature}_1 + w_2 * \text{feature}_2 \dots w_k * \text{feature}_k$

$$\begin{aligned} \text{target} = & 2.066645 + (0.846924 * \text{MedInc}) + (0.118799 * \text{HouseAge}) + \\ & (-0.299525 * \text{AveRooms}) + (0.348115 * \text{AveBedrms}) + (0.000040 * \text{Population}) + \\ & (-0.041724 * \text{AveOccup}) + (-0.894496 * \text{Latitude}) + (-0.870096 * \text{Longitude}) \end{aligned}$$

2. The MSE and RMSE of the training data, as well as the test data

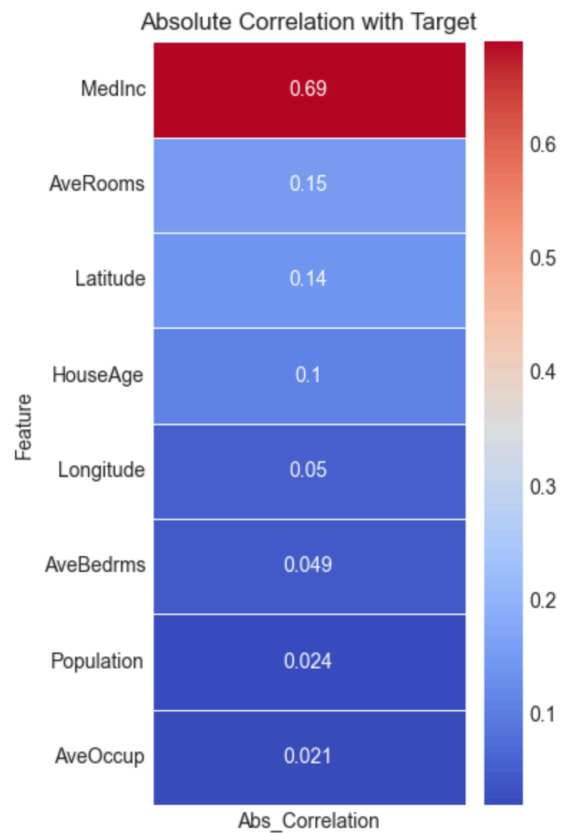
Model	Train RMSE	Test RMSE	Train MSE	Test MSE
Linear Regression	0.720799	0.7332	0.529148	0.529868

3. Feature importance

- A. The features ranked using correlation (descending order based on absolute value)

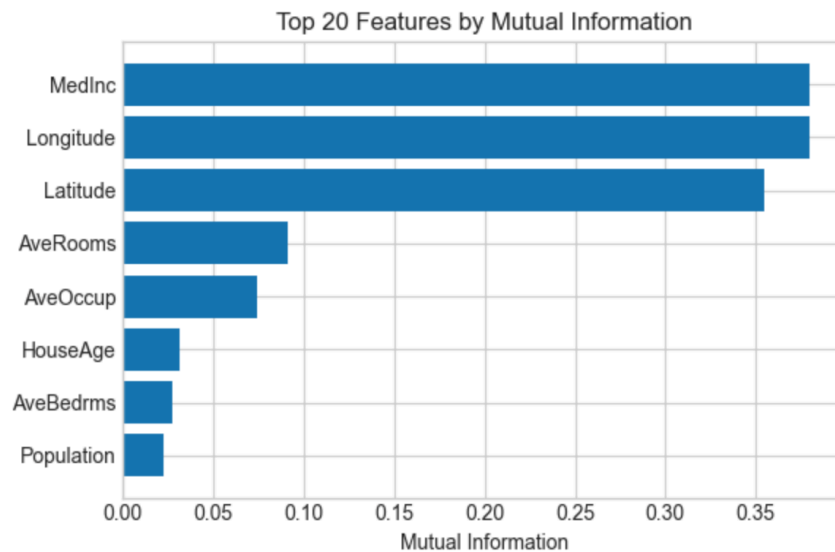
Here are the features ranked using correlation with the target in descending order based on the absolute value

Feature	Correlation	Abs_Correlation
MedInc	0.689075	0.689075
AveRooms	0.151918	0.151918
Latitude	-0.140136	0.140136
HouseAge	0.104243	0.104243
Longitude	-0.050082	0.050082
AveBedrms	-0.048867	0.048867
Population	-0.023560	0.023560
AveOccup	-0.020558	0.020558



B. The features ranked using Mutual Information (descending order based on absolute value)

Feature	Mutual_Information
MedInc	0.380122
Longitude	0.379757
Latitude	0.354531
AveRooms	0.091331
AveOccup	0.073808
HouseAge	0.031459
AveBedrms	0.027574
Population	0.022005



C. Do the two rankings agree?

According to both rankings, MedInc has the highest value. So, in predicting the target “MedHouseVal”, the feature “MedINC” is the most dominant predictor in the dataset according to both rankings. However, in both plots, we observe that the order of the other features after MedInc differs significantly between the correlation and mutual information rankings. For example, the second most important feature, according to the absolute value of the correlation ranking, is AveRooms, whereas it is Longitude according to the mutual information ranking.

D. Which features are consistently important?

According to both rankings (based on correlation with the target and mutual information), the feature “MedInc” has the highest value, indicating it is consistently important.

4. Coefficient interpretation

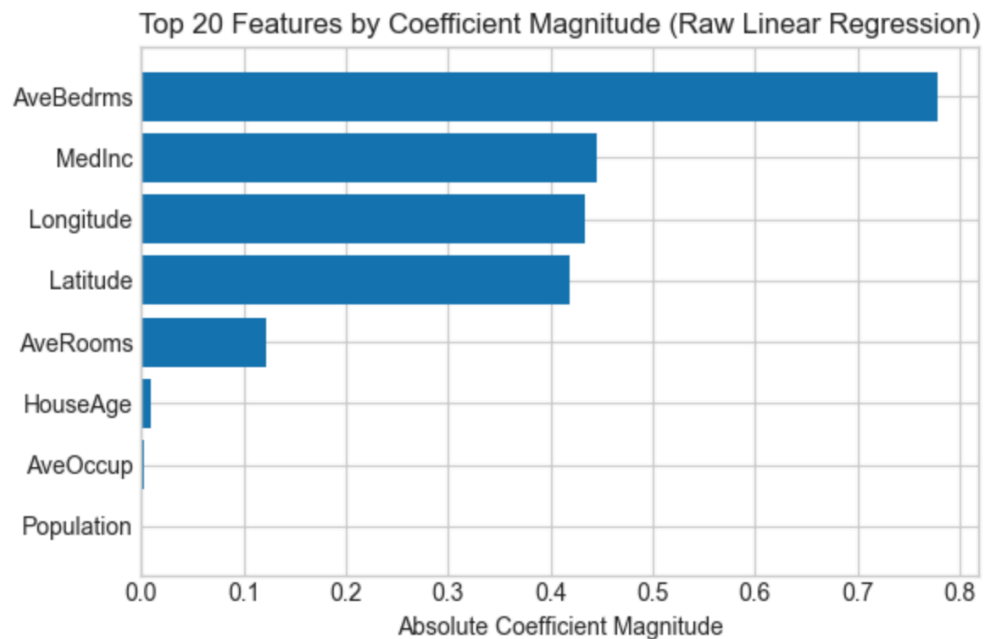
A. Which feature has the largest impact?

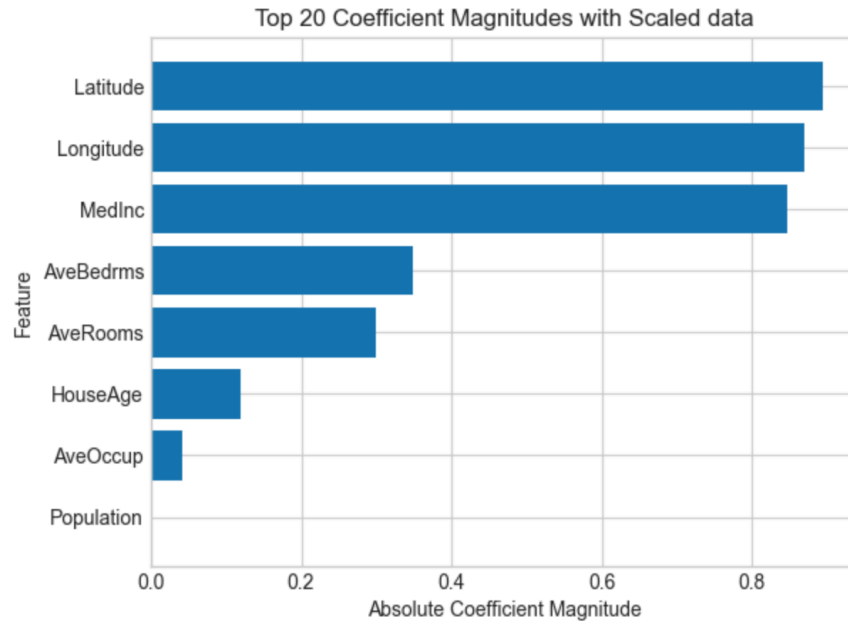
The largest absolute coefficient is Latitude ($|-0.894496|$). So, latitude has the largest impact on the target.

B. Which feature surprised you?

The Latitude and Longitude features surprised me because they are larger than the MedInc. In general, we expect MedInc or median income to be the dominant predictor of housing value. But the model suggests that geographic location, such as Latitude and Longitude, has an even stronger effect than income.

C. Does coefficient magnitude depend on scaling? (may answer by experiment) Why do you think that is the case?





As shown in the two plots above, the magnitude of the coefficients depends strongly on the scaling data. The magnitude of the coefficients differs significantly between scaled and raw data.

According to the raw data, the most significant feature is AveBedrms, as its coefficient is the largest among the other features. However, according to the scaled data, the most significant feature is Latitude, as its coefficient magnitude is the largest among the other features.

This is expected because feature comparisons, depending on the magnitude of the coefficients, can be misleading when using unscaled/raw data, as a feature with a larger numeric range may appear more important.

- Problem 3 – Ridge and Lasso Regularization

1. How does the MSE and RMSE differ between Original Linear Regression (OLR), Ridge regression and Lasso regression, for training data and for test data? Show your answer in a table format to clearly view the pattern.

Model	Train_RMSE	Test_RMSE	Train_MSE	Test_MSE	Train_R2	Test_R2
OLS	0.720799	0.733200	0.529148	0.529868	0.609603	0.596705
Lasso_alpha_0.01	0.724630	0.732121	0.532489	0.533153	0.605442	0.597890
Ridge_alpha_10	0.720808	0.733053	0.529100	0.529839	0.609593	0.596866

Which method performs best in training data?

Based on the **training data performance**, the method that performs best is **Ordinary Least Squares (OLS)**, although the difference compared to Ridge is extremely small.

Looking at the training metrics:

- **OLS** has the lowest Train RMSE (0.720799) and a high Train R^2 (0.609603).
- **Ridge ($\alpha = 10$)** performs almost identically, with Train RMSE = 0.720808 and Train $R^2 = 0.609593$, differing only at the fourth decimal place.
- **Lasso ($\alpha = 0.01$)** performs slightly worse on training data, with a higher Train RMSE (0.724630) and lower Train R^2 (0.605442), which is expected because Lasso applies stronger regularization and shrinks coefficients.

Therefore, **OLS performs best on the training data** because it minimizes training error without any regularization constraints. Ridge performs nearly the same due to mild coefficient shrinkage, while Lasso sacrifices some training accuracy in exchange for better regularization and potential generalization benefits.

Which method performs best in test data?

Based on the **test data performance**, the method that performs best is **Lasso regression with $\alpha = 0.01$** .

Looking at the test metrics:

- **Lasso ($\alpha = 0.01$)** has the **lowest Test RMSE (0.732121)** and the **highest Test R^2 (0.597890)**, indicating the best predictive accuracy on unseen data.
- **Ridge ($\alpha = 10$)** performs very similarly but slightly worse, with Test RMSE = 0.733053 and Test $R^2 = 0.596866$.
- **OLS** has the highest Test RMSE (0.733200) and the lowest Test R^2 (0.596705) among the three models.

Although the differences are small, Lasso provides the best generalization performance because its regularization reduces overfitting while retaining predictive power. Therefore, **Lasso with $\alpha = 0.01$ performs best on the test dataset.**

2. Does Ridge regression shrink coefficients? Show the values of the coefficients to support your answer.

df_ridge_alpha_results_shrinking										
	Model	Intercept	Num_Features	NonZero_Coeff	Killed_Coeff	L2_Norm	L1_Norm	Mean_Abs_Coeff	Max_Abs_Coeff	Alpha
0	Ridge_alpha_0.001	2.066645	8	8	0	1.581524	3.419715	0.427464	0.894495	0.001
1	Ridge_alpha_0.01	2.066645	8	8	0	1.581513	3.419694	0.427462	0.894486	0.010
2	Ridge_alpha_0.1	2.066645	8	8	0	1.581408	3.419483	0.427435	0.894405	0.100
3	Ridge_alpha_1	2.066645	8	8	0	1.580358	3.417375	0.427172	0.893590	1.000
4	Ridge_alpha_10	2.066645	8	8	0	1.569969	3.396472	0.424559	0.885535	10.000

Yes, **Ridge regression does shrink coefficients**, but it does so **gradually rather than eliminating them**. The values in your table clearly support this behavior.

Across all Ridge models, the number of nonzero coefficients remains **8**, and the number of killed coefficients is **0**, meaning Ridge does not set coefficients exactly to zero. However, as the regularization parameter α increases, the overall magnitude of the coefficients steadily decreases.

This shrinkage can be observed from several metrics:

- The **L2 norm** decreases from **1.581524** ($\alpha = 0.001$) to **1.569969** ($\alpha = 10$).
- The **L1 norm** decreases from **3.419715** to **3.396472** as α increases.
- The **L1 norm** decreases from **3.419715** to **3.396472** as α increases.
- The **mean absolute coefficient** drops from **0.427464** to **0.424559**.

These consistent reductions show that Ridge regression penalizes large coefficients and pulls them closer to zero as the regularization strength increases. Ridge regression **shrinks all coefficients smoothly while keeping every predictor in the model**. Therefore, the coefficient norms and magnitudes in the table confirm that Ridge regression performs coefficient shrinkage without feature elimination.

3. Does Lasso regression reduce features by zero out the coefficients? If so, which features were zeroed out?

Model	Num_Features	NonZero_Coeff	Killed_Coeff	L2_Norm	L1_Norm	Mean_Abs_Coeff	Max_Abs_Coeff	Alpha
Lasso_alpha_0.001	8	7	1	1.560854	3.364320	0.420540	0.883839	0.001
Lasso_alpha_0.01	8	7	1	1.382722	2.866020	0.358252	0.793322	0.010
Lasso_alpha_0.1	8	3	5	0.713663	0.817908	0.102238	0.706193	0.100
Lasso_alpha_1	8	0	8	0.000000	0.000000	0.000000	0.000000	1.000
Lasso_alpha_10	8	0	8	0.000000	0.000000	0.000000	0.000000	10.000

Yes, **Lasso regression reduces features by setting some coefficients exactly to zero**, which performs automatic feature selection. The values in the table clearly support this behavior.

As the regularization parameter α increases, the number of **nonzero coefficients decreases** while the number of **killed coefficients increases**. For example, when $\alpha = 0.001$ and $\alpha = 0.01$, the model retains 7 out of 8 features and zeroes out 1 feature. When α increases to 0.1, only 3 features remain active, and 5 features are eliminated. For larger values ($\alpha = 1$ and $\alpha = 10$), all coefficients become zero, meaning the model removes every feature and predicts only the intercept.

Lasso ($\alpha = 0.01$) zeroed out exactly one feature: Population.

4. How to identify the best a value for Ridge and Lasso regression for this data set?

Model	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_R2	Test_R2	Alpha
Ridge_alpha_0.001	0.720799	0.733200	0.529148	0.529868	0.609603	0.596705	0.001
Ridge_alpha_0.01	0.720799	0.733199	0.529148	0.529868	0.609603	0.596705	0.010
Ridge_alpha_0.1	0.720799	0.733198	0.529148	0.529868	0.609603	0.596706	0.100
Ridge_alpha_1	0.720799	0.733184	0.529143	0.529864	0.609603	0.596722	1.000
Ridge_alpha_10	0.720808	0.733053	0.529100	0.529839	0.609593	0.596866	10.000

The above tables suggest that the dataset is not very sensitive to ridge regularization. There may not be strong multicollinearity, and the coefficients are already stable. In such cases, the ridge barely changes performance. The best alpha is alpha=10 because the lowest Test RMSE, the highest Test R^2 , and slightly better generalization.

Model	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_R2	Test_R2	Alpha
Lasso_alpha_0.001	0.720838	0.732693	0.529205	0.529927	0.609561	0.597261	0.001
Lasso_alpha_0.01	0.724630	0.732121	0.532489	0.533153	0.605442	0.597890	0.010
Lasso_alpha_0.1	0.820885	0.823665	0.619617	0.621466	0.493659	0.491045	0.100
Lasso_alpha_1	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	1.000
Lasso_alpha_10	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	10.000

Based on the model performance table, the optimal alpha for the Lasso regression model is **0.01**. This value produces the lowest Test RMSE (0.732121) and the highest Test R² (0.597890), indicating the best generalization performance on unseen data. When alpha is set to 0.001, the model performs similarly but slightly worse, suggesting that very weak regularization behaves almost like ordinary least squares. However, when alpha increases to 0.1, performance deteriorates noticeably, with a sharp increase in Test RMSE and a drop in Test R², indicating underfitting due to excessive shrinkage. For alpha values of 1 and 10, all coefficients are effectively driven to zero, resulting in a model that predicts only the intercept and yields near-zero explanatory power. Therefore, alpha = 0.01 provides the best balance between reducing overfitting through regularization and maintaining predictive accuracy.

5. What are the values if one considers a value between 10^{-3} to 10^3 in log scale?

	Model	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_R2	Test_R2	Alpha
0	Lasso_alpha_0.00100	0.720838	0.732693	0.529205	0.529927	0.609561	0.597261	0.001000
1	Lasso_alpha_0.00207	0.720964	0.732250	0.529339	0.530047	0.609425	0.597749	0.002069
2	Lasso_alpha_0.00428	0.721503	0.731652	0.529835	0.530526	0.608840	0.598406	0.004281
3	Lasso_alpha_0.00886	0.723807	0.731795	0.531806	0.532441	0.606338	0.598248	0.008859
4	Lasso_alpha_0.01833	0.733590	0.737984	0.539934	0.540758	0.595625	0.591424	0.018330
5	Lasso_alpha_0.03793	0.759681	0.761849	0.561432	0.562072	0.566349	0.564572	0.037927
6	Lasso_alpha_0.07848	0.813559	0.815888	0.612305	0.614114	0.502657	0.500610	0.078476
7	Lasso_alpha_0.16238	0.844196	0.847743	0.642273	0.643925	0.464493	0.460852	0.162378
8	Lasso_alpha_0.33598	0.901001	0.904316	0.697984	0.699317	0.390002	0.386494	0.335982
9	Lasso_alpha_0.69519	1.087296	1.088804	0.858745	0.860052	0.111672	0.110638	0.695193
10	Lasso_alpha_1.43845	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	1.438450
11	Lasso_alpha_2.97635	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	2.976351
12	Lasso_alpha_6.15848	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	6.158482
13	Lasso_alpha_12.74275	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	12.742750
14	Lasso_alpha_26.36651	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	26.366509
15	Lasso_alpha_54.55595	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	54.555948
16	Lasso_alpha_112.88379	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	112.883789
17	Lasso_alpha_233.57215	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	233.572147
18	Lasso_alpha_483.29302	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	483.293024
19	Lasso_alpha_1000.00000	1.153615	1.154558	0.910782	0.912475	0.000000	-0.000025	1000.000000

Based on the log-scale search of alpha values between 10^{-3} and 10^3 , the optimal Lasso regularization parameter is approximately **0.00428**. This value yields the lowest Test RMSE (0.731652) and the highest Test R^2 (0.598406), indicating the best predictive performance on unseen data. While very small alpha values such as 0.001 and 0.00207 produce similar results, $\alpha = 0.00428$ provides a slight but measurable improvement in generalization. As alpha increases beyond roughly 0.018, model performance begins to deteriorate, indicating increasing underfitting due to stronger shrinkage. For alpha values near 1 or above, all coefficients converge to zero, causing the model to collapse to the intercept-only solution with essentially no explanatory power. Therefore, $\alpha \approx 0.0043$ represents the best balance between regularization strength and predictive accuracy in this log-scale search.

	Model	Train_RMSE	Test_RMSE	Train_MSE	Test_MSE	Train_R2	Test_R2	Alpha
	Ridge_alpha_0.00100	0.720799	0.733200	0.529148	0.529868	0.609603	0.596705	0.001000
	Ridge_alpha_0.00207	0.720799	0.733200	0.529148	0.529868	0.609603	0.596705	0.002069
	Ridge_alpha_0.00428	0.720799	0.733199	0.529148	0.529868	0.609603	0.596705	0.004281
	Ridge_alpha_0.00886	0.720799	0.733199	0.529148	0.529868	0.609603	0.596705	0.008859
	Ridge_alpha_0.01833	0.720799	0.733199	0.529148	0.529868	0.609603	0.596705	0.018330
	Ridge_alpha_0.03793	0.720799	0.733199	0.529148	0.529868	0.609603	0.596705	0.037927
	Ridge_alpha_0.07848	0.720799	0.733198	0.529148	0.529868	0.609603	0.596706	0.078476
	Ridge_alpha_0.16238	0.720799	0.733197	0.529148	0.529867	0.609603	0.596707	0.162378
	Ridge_alpha_0.33598	0.720799	0.733194	0.529147	0.529866	0.609603	0.596710	0.335982
	Ridge_alpha_0.69519	0.720799	0.733189	0.529145	0.529865	0.609603	0.596717	0.695193
	Ridge_alpha_1.43845	0.720800	0.733177	0.529141	0.529862	0.609603	0.596729	1.438450
	Ridge_alpha_2.97635	0.720800	0.733154	0.529132	0.529857	0.609602	0.596755	2.976351
	Ridge_alpha_6.15848	0.720803	0.733107	0.529117	0.529848	0.609599	0.596807	6.158482
	Ridge_alpha_12.74275	0.720814	0.733016	0.529090	0.529835	0.609587	0.596906	12.742750
	Ridge_alpha_26.36651	0.720861	0.732856	0.529059	0.529824	0.609536	0.597083	26.366509
	Ridge_alpha_54.55595	0.721049	0.732631	0.529104	0.529872	0.609332	0.597330	54.555948
	Ridge_alpha_112.88379	0.721760	0.732551	0.529522	0.530287	0.608562	0.597418	112.883789
	Ridge_alpha_233.57215	0.724161	0.733567	0.531362	0.532299	0.605952	0.596301	233.572147
	Ridge_alpha_483.29302	0.730898	0.738256	0.537041	0.538395	0.598587	0.591123	483.293024
	Ridge_alpha_1000.00000	0.745467	0.750568	0.550902	0.551919	0.582424	0.577372	1000.000000

Based on the log-scale evaluation of Ridge regression models, the optimal regularization parameter is $\alpha = \mathbf{112.88}$. This value yields the lowest Test RMSE (0.732551) and the highest Test R^2 (0.597418), indicating the best predictive performance on unseen data. For small alpha values, Ridge behaves similarly to ordinary least squares because the penalty is weak, resulting in nearly identical performance. As alpha increases, moderate regularization slightly improves generalization by shrinking coefficients and reducing variance. However, when alpha becomes too large (greater than about 200), the model begins to underfit, leading to increasing prediction error. Therefore, $\alpha = 112.88$ provides the best balance between bias and variance for the Ridge regression model.

Assume the Standard-error loss for linear regression

$$J(w_0, w_1) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2, \quad y_i = w_0 + w_1 x_i, \quad m=3$$

$$\text{Gradients: } g_0 = \frac{\partial J}{\partial w_0} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i),$$

$$g_1 = \frac{\partial J}{\partial w_1} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_i$$

Update rule with learning rate $\lambda = 0.1$:

$$w_0 \leftarrow w_0 - \lambda g_0$$

$$w_1 \leftarrow w_1 - \lambda g_1$$

$$\text{Data: } (x, y) = \{(0, 2), (1, 5), (2, 8)\}$$

Iteration 1 ~~Start~~

$$(\text{start } w_0 = 0, w_1 = 0)$$

$$\text{Prediction: } \hat{y} = [0, 0, 0]$$

$$\text{Errors: } e = \hat{y} - y = [-2, -5, -8]$$

Loss:

$$J = \frac{1}{6} (4 + 25 + 64) \\ = \frac{93}{6} = 15.5$$

Gradients:

$$g_0 = \frac{1}{3} (-2 - 5 - 8) = \frac{-15}{3} = -5$$

$$g_1 = \frac{1}{3} ((-2) \cdot 0 + (-5) \cdot (1) + (-8) \cdot 2) \\ = \frac{-21}{3} = -7$$

update:

$$w_0 = 0 - 0.1(-5) = 0.5$$

$$w_1 = 0 - (0.1)(-7) = 0.7$$

new loss (after update):

$$J = 8.983 \rightarrow \text{decreased (15.5 to 8.983)}$$

Iteration 2 (Start $w_0 = 0.5, w_1 = 0.7$)

Prediction:

$$\hat{y} = [0.5, 1.2, 1.9]$$

Errors: $e = [-1.5, -3.8, -6.1]$

Loss:

$$J = \frac{1}{6} ((1.5)^2 + (3.8)^2 + (6.1)^2)$$

$$= \frac{1}{6} (2.25 + 14.44 + 37.21)$$

$$= 8.983$$

Gradient:

$$g_0 = \frac{1}{3} (-1.5 - 3.8 - 6.1)$$

$$= \frac{-11.4}{3} = -3.8$$

$$g_1 = \frac{1}{3} ((-1.5) \cdot 0 + (-3.8) \cdot 1 + (-6.1) \cdot 2)$$

$$= \frac{-16.0}{3} = -5.33$$

update:-

$$w_0 = 0.5 - (0.1)(-3.8) = 0.88$$

$$w_1 = 0.7 - (0.1)(-5.33) = 1.233$$

New loss: $J = 5.206793$

So it is decreased from 8.9833 to 5.206793

Iteration 3 (Start $w_0 = 0.88$, $w_1 = 1.233$)

predictions:-

$$\hat{y} = [0.88, 2.1133, 3.34]$$

Error:-

$$e = [-1.12, -2.88667, -4.6533]$$

Loss:

$$\begin{aligned} J &= \frac{1}{6} [(1.12)^2 + (2.88667)^2 + (4.6533)^2] \\ &= 5.206793 \end{aligned}$$

Gradients:

$$g_0 = \frac{1}{3} (-1.12 - 2.88667 - 4.6533) \\ = -2.88667$$

$$g_1 = \frac{1}{3} ((-1.12) \cdot 0 + (-2.88667) \cdot 1 + (-4.6533) \cdot 2) \\ = -4.06444$$

Update:

$$w_0 = 0.88 - 0.1 (-2.88667) \\ = 1.168667$$

$$w_1 = 1.2333 - 0.1 (-4.06444) \\ = 1.639778$$

New loss: $J = 3.018193$

which is decreased from 5.206793